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| **Fiscal Year:** | FY 2010 | **Task Last Updated:** | FY 11/16/2009 |
| **PI Name:** | Dunn-Rankin, Derek Ph.D. | | |
| **Project Title:** | Applications of Electric Fields in Microgravity Combustion Science (NNX07AB55A) | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | COMBUSTION SCIENCE | | |
| **Program/Discipline--Element/Subdiscipline:** | COMBUSTION SCIENCE--Droplets, sprays, particles, dust clouds, nanoparticles | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | ddunnran@uci.edu | **Fax:** | FY |
| **PI Organization Type:** | UNIVERSITY | **Phone:** | 949-824-8745 |
| **Organization Name:** | University of California - Irvine | | |
| **PI Address 1:** | Department of Mechanical & Aerospace Engineering | | |
| **PI Address 2:** | 4200 Engineering Gateway Bldg, EG3224 | | |
| **PI Web Page:** | | | |
| **City:** | Irvine | **State:** | CA |
| **Zip Code:** | 92697-3975 | **Congressional District:** | 48 |
| **Comments:** | | | |
| **Project Type:** | Flight | **Solicitation / Funding Source:** | 2001 Physical Sciences 01-OBPR-08 |
| **Start Date:** | 10/01/2006 | **End Date:** | 09/30/2011 |
| **No. of Post Docs:** | | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | 1 | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | 1 | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | 1 | **Monitoring Center:** | NASA GRC |
| **Contact Monitor:** | Stocker, Dennis P | **Contact Phone:** | 216-433-2166 |
| **Contact Email:** | dennis.p.stocker@nasa.gov | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | Flight definition.<br>NOTE: Extended to 9/30/2011 per NSSC info (9/2009)<br>NOTE: NEW grant, resulting from grants reinstatement directive in 8/2006, per D. Stocker/GRC (10/07) | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Weinberg, Felix ( Imperial College, London, England )<br>Yuan, Zeng-Guang ( National Center for Space Exploration Research, NASA Glenn Research Center ) | | |
| **Grant/Contract No.:** | NNX07AB55A | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

**Task Description:**

This joint project between the University of California, Irvine and Imperial College, London, with support from researchers at NASA Glenn Research Center, explores the use of large electric fields in combustion for the purpose of improving performance of combustion-based energy conversion systems. Electric fields have been shown to affect flame stability and soot formation; using the flames electrical response as a sensor, it is also possible to describe flame character. Recently, the project has focused on continued electric field testing on a coflow burner for both methane and ethylene, which is proposed as a potential ACME spaceflight experimental platform, and fundamental studies involving partially premixing air in the fuel stream of a jet flame. Current studies include:
1) High resolution voltage-current characteristics under steady DC-fields and the dynamic response of flames to such fields. Testing includes, measuring the flame's electrical response and voltage-current characteristic under various fuel and air co-flow conditions, including testing at reduced gravity using NASA Glenn 2.2-s Drop Tower.

2) Experiments and numerical studies of CO and H2 concentrations in flames with varying levels of aeration.

Task Description:

The overall goal of this project is to understand the behavior of chemi-ionization in flames and to demonstrate the use of this understanding to manipulate combustion in beneficial ways (including enhancing flame stability and reducing flame soot). The project uses detailed measurements of electrical properties of flames in several ways.

First, we can create a neutral wind (ion driven convection) locally at the flame front; this is one of the rare and important action-at-a-distance effects on flames that can be used as a localized actuator. The effect of the wind has been demonstrated with a candle flame in parabolic flight based microgravity and in a spherical cage drop tower, but there has been no systematic study of the ion wind effect in the absence of the confounding effects of the approximately equal influence of buoyancy.

Second, we can use electric fields to affect soot formation and transport – ions and soot are closely linked because there are shared chemical pathways and because soot particles are charged by the flame ions. It is possible, therefore, to modify soot formation and to use charges attached to soot to affect soot transport. By adjusting the soot trajectory through the reaction zone it is possible to affect the oxidation of soot to prevent its eventual release or to modify the residence time of soot in different zones of the flame. It is also possible to use the neutral ion wind to change the mixing behavior of the flame and reduce soot in this way.

Finally, we can use the electrical properties to characterize the flame – there is a quantifiable relationship between the carbon content of a fuel and the ion current it can produce at saturation. This relationship is already exploited in a typical flame ionization detector (FID) analyzer for hydrocarbon emission where the carbon is burned in a hydrogen flame to create an ion current in proportion to the carbon influx. In direct flame interrogation, however, there are several confounding variables that can affect the ion current, including flame temperature (which generally affects the O atom concentration) and the reaction zone volume. Flame temperature can in turn be affected by levels of premixing, diffusion rates, fuel composition, and radiative loss. It is possible, therefore, to determine flame character from the ionization curve response.

Task Significance:

Understanding electric field interactions with flames is central to combustion control, particularly in situations near limit operation of the flame where small effects are amplified dramatically. Electric fields have been shown to modify burning rates as well as to enhance and reduce flame propagation even to the point of flame extinguishment. Our experiments provide a unique ability to make simultaneous measurements of both physical and electrical properties of flames.

**Rationale for HRP Directed Research:**

**Research Impact/Earth Benefits:**

Several of the applications we explore have potential uses in earth gravity. Understanding fundamental electrical properties of flames have contributions to earth-based combustion, particularly as regards soot formation and control. Ion current is also a very useful diagnostic tool and sensor for combustion, capable of reflecting flame character and behavior for next generation power generation systems such as homogeneous charge compression ignition engines. Finally, there are important potential applications in combustion synthesis of nanoparticles where ion formation pathways are important.

Experimental work was accomplished using a coflow burner. The burner consists of two stainless steel plenums threaded together. Fuel is mixed, if desired, with nitrogen in the bottom plenum and passed through a stainless steel tube (2.13 mm ID). The flow at the exit is fully developed. Air is introduced in the top chamber, passing through beads, and a honeycomb mesh to create a uniform flow at the exit. A variable high potential supply creates an electric field between the burner and mesh electrode. Field strength between the electrodes is not linear due to the flame's space charge; however, the field can be approximated as E = V/d, where V is the applied potential and d is the distance between electrodes. The above approximation is used in describing the relative field strength used for different experiments. Ion current is determined using Ohm's law and the potential drop across a shunt resistor placed between the electrically isolated burner and the building ground.
An interesting and new finding this year was that some voltages lead to an oscillating flame. The term "oscillation" is used as opposed to flicker because of the wide behavioral range of motion. Tip and bulk flickering, are described as the flame top oscillating or elongating periodically; and the flame separating from the top of the main body, respectively. Depending on conditions, ion current oscillations due to a static field can involve the movement of the entire flame body or no movement at all. As an example, the transient ion current for a CH4 flame (23.2 mg/min) in a coflow, at an applied field of -2.15 kV/cm, shows a ~43Hz oscillation. The ion current fluctuation is not obvious in the flame luminosity, shape or position. Initial PMT measurements are able to capture the fluctuations, which match in frequency the ion current fluctuation, albeit with a small phase lag (~5ms). The period of the peak oscillation is ˜23ms. The response time of the flame is ~100ms, assuming a 1cm reaction zone and a burned gas density that is 15% of ambient. The time scale suggests that ion current oscillations are too fast to cause a noticeable movement or disruption of the flame.

Ethylene studies -- Soot is known to be charged and will influence ion current measurements. Ethylene is used because of its simple structure and high sooting propensity. A unique ion current response is observed with ethylene as the fuel

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| **Task Progress:** | source. Luminous soot in the flame is strongly affected by the applied field. At a flow rate of 35.4 cc/min C2H4, electric fields destabilize the flame with the oscillation of luminous soot dominating the flame's movement. Near saturation, the flame begins to oscillate at ~23Hz. Higher applied fields distort the flame shape, although not affecting the frequency, resulting in heavy soot deposits on the honeycomb annulus. Nitrogen dilution of the fuel muted the flame distortion. Images show a gradual reduction of luminous soot with applied field. The most interesting finding is in the absolute ion current measurements for an ethylene flame with varies levels of nitrogen dilution. The measured ion current per unit carbon increases with nitrogen dilution. Premixing ethylene with nitrogen reduces combustion temperatures and the amount of luminous soot. The high thermal ionization potential of soot, due to its low work function, complicate ion current measurements by introducing additional electron emitters. The role of ions in soot formation is an unresolved question; however, soot extinction measurements would aid the in the understanding of how an electric field affects soot luminosity.<br><br>Partially premixed studies – Professor Felix Weinberg at Imperial College is currently studying the effects of partial aeration on the structure of flames. Previous work involved CO and H2 sampling by means of a quartz microprobe mounted on an x-y-z translation stage for regions bounded by the I. D. of the burner mouth. The data showed large concentrations of CO and H2 inside the flame boundary. As the flame base overhangs that diameter to a substantial extent, sampling was extended to larger radii. CO concentrations for 75% air admixture as contour lines at 0.5% intervals were measured. The maximum at the core of the torus is 5.5 volume %. Similar structures are seen in CHEMKIN simulations.<br><br>A surprising result was the appearance of CO in flames in the absence of air admixture. Moreover, some of it appeared well within the burner mouth. The implication is that we never have a pure diffusion flame and we suspected air entrainment, possibly due to convection currents occasioned by the elevated burner temperature (> 300oC) to be the cause. To ascertain the extent of penetration below the plane of the burner mouth, it was necessary to replace the normally horizontal sampling capillary by a vertical probe. Comparing CO and H2 percentages measured on the burner axis with different probes orientations show measurable concentrations of CO and H2 occur at least 1mm and 3mm, respectively, below the plane of the burner mouth. This is for the extreme case of no aeration. The penetration decreases with increasing air addition – and hence flow velocity - until at 75% air there is no CO below the rim and H2 at 1mm below is reduced to one tenth of the value compared to the pure fuel case. Whilst the results close to the burner mouth and well above it are very similar for the two orientations, the maxima are somewhat greater and displaced upward in the case of the vertical probe.<br><br>CO contour maps suggest that no CO would emerge into the atmosphere if the flame reaction proceeded to completion normally. However, if the reaction was arrested by quenching before completion, (particularly, if occurring close to the peak of CO concentration) the consequent CO escape might pose a serious hazard. In order to examine this possibility, we compared CO volumetric flow rates of a cooled and uncooled burner for 75% air addition; they show similar results, albeit somewhat displaced with respect to each other. The average CO emitted is around 2.6ml/min. That appears close to the maximum attainable, in that the curves suggest a fall in CO, had it been possible to lower the quenching surface further without extinguishing the flame. The highest CO emission rate measured is about 4ml/min. This is close to 10% of the CH4 fed into the burner and clearly poses a potentially serious hazard, as will be apparent if the CH4 feed is expressed in terms kilowatts for a practical boiler.<br><br>We continue to evaluate the potential CO emission and how electric fields can be used to mitigate these releases. |

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| **Bibliography Type:** | Description: (Last Updated: 11/26/2025) |
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